



Microstructure and properties of FeCoNiCrX (X=Mn, Al) high-entropy alloy coatings

Zhiyi Rong^a, Chaohui Wang^{a,*}, You Wang^b, Meiling Dong^a, Yuan You^a, Jianing Wang^a, Huinan Liu^a, Jiaqi Liu^a, Yuhang Wang^a, Zhongyu Zhu^a

^a School of Materials Science and Engineering, Qiqihar University, Qiqihar 161006, PR China

^b School of Materials Science and Engineering, Harbin Institute of Technology, Harbin 150001, PR China

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ABSTRACT

FeCoNiCrX (X=Mn, Al) high-entropy alloy (HEA) coatings were prepared onto a copper substrate by electrodeposition method. The microstructure, hardness, wear, and corrosion of the as-prepared HEA coatings were studied in this paper. The as-prepared HEA coatings exhibited simple crystal structured solid-solution phases, face-centered cubic (FCC) phase. It was shown in the SEM photographs that the surfaces of the as-prepared HEA coatings are uniform and smoothness. Results show that the microhardness of the as-prepared HEA coatings was 4 times higher than that of substrate, and coefficient of friction and wear volume of HEA coatings were lower than that of substrate in friction and wear experiments. The corrosion resistance of the HEA coatings is better than that of the substrate in 3.5 % NaCl solution.

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1. Introduction

High-entropy material is a new class of multi-principal element alloys materials that have emerged in recent years, consisting of multiple elements in equal or near-equal amount. The traditional concept of material design has been updated. High-entropy materials display various organizational and performance that are distinct from ordinary materials, notably high-entropy alloys (HEA) [1–5] and high-entropy ceramics [6–10], due to their unique crystal structure. The HEA was initially introduced by Yeh et al. [1], which defined as a new alloy system with least five principal elements, each of which has an atomic concentration between 5 % and 35 %. This concept breaks away from the traditional design of alloys based on one or two elements and opens up a new direction for the development of alloys. HEAs offer significant advantages due to their high thermodynamic entropy, structural lattice distortion, kinetic hysteresis-diffusion effects and "cocktail" of properties. Their unique mechanical, electrical and magnetic properties have become a new focus of materials research.

In recent years, there have been many important researches on high-entropy alloys, and many scholars used high-entropy alloys for surface modification [11–13]. High-entropy alloys have been developed, because of their excellent mechanical properties and corrosion

resistance. As a result, the use of low-cost high-entropy alloy coatings instead of bulk high-entropy alloys will reduce the cost and increase the scope of application. By adjusting the alloy constituents, high entropy alloys with the desired properties can be obtained. The atomic radius and electronegativity of Fe, Co, Ni, and Cr are similar, easy to form a single FCC structure. The presence of metal oxides in high entropy alloys has a great influence on the properties of materials, for example, there is such class of materials as the complex iron oxides that are also promising for practical use as electronic materials [14–19]. The corrosion resistance of the as-prepared HEA coatings can be improved by adding some passivating elements such as Co, Ni, Cr, Al, and Mn, furthermore, lattice distortion caused by the entry of modified elements into the lattice improves the hardness and strength of the coatings. Currently, there are many surface modification methods to prepare HEA coatings, for example, magnetron sputtering [20,21], spraying [22], laser cladding [23,24], and so on. Even if the material composition is the same, the coatings performance attained through various preparation processes is markedly different. The electrodeposition method has the advantages of low cost, simple operation, high deposition efficiency and friendly environment, which can be applied to prepare alloy coatings [25]. Therefore, the high-entropy alloy coatings can be obtained by electrodeposition to enhance the corrosion resistance and mechanical properties of the substrate. In this paper, high-entropy alloy coatings were prepared on copper substrate by

* Correspondence to: Qiqihar University, Qiqihar 161006, PR China.

E-mail addresses: R799rzy@163.com (Z. Rong), wch800209@126.com (C. Wang).

Table 1
Composition bath of the FeCoNiCr series HEA coatings.

Bath composition/Sample	FeCoNiCr	FeCoNiCrAl	FeCoNiCrMn
FeSO ₄ ·7H ₂ O	2	1.8	1.6
CoSO ₄ ·7H ₂ O	4	3.2	2.8
NiSO ₄ ·7H ₂ O	20	10	8
Cr ₂ (SO ₄) ₃ ·6H ₂ O	240	120	180
MnSO ₄ ·H ₂ O	–	–	100
Al ₂ (SO ₄) ₃ ·18H ₂ O	–	80	–
H ₃ BO ₃	25	40	25
KCl	75	90	75
C ₆ H ₈ O ₇ ·H ₂ O	92	92	92
C ₆ H ₅ Na ₃ O ₇ ·2H ₂ O	48	48	48
NaH ₂ PO ₂ ·H ₂ O	1	1	1

electrodeposition using Fe, Co, Ni, Cr, and Mn/Al as the main alloy systems, and their structures and properties were researched.

2. Experimental

2.1. Electrodeposition

Copper substrate (10 mm × 10 mm) and graphite plate with the same exposed area were used as the cathode and anode, respectively. The copper substrate was abraded with different SiC abrasive papers for achieving smooth surface, followed by ultrasonically cleaning in acetone. This process facilitates the adhesive bonding of the coatings to the substrate, and also prevent the rapid discharge of current during electrodeposition, which affects the plating properties. The substrate was chemically cleaned by immersing in chemical degreasing solution for 10 min. After that, the substrate was removed from the degreasing solution, cleaned with distilled water, and then immersed in an acidic polishing solution. Prior to coatings, the copper substrate was ultrasonically cleaned in acetone, activated in 5 wt% HCl at room temperature for 10 s, washed in distilled water and then placed immediately in the electrodeposition bath as cathode.

In this experiment, FeCoNiCr, FeCoNiCrMn, FeCoNiCrAl high-entropy alloy coatings were prepared by electrodeposition under acidic conditions with the same process parameters. The composition of the plating solution is shown in Table 1. The plating process parameters are: pH 1.5, temperature 40 °C, current density 20 A/dm², plating time 30 min, stirring speed 250 r/min. All reagents used in the experiments were of analytical grade and solutions were prepared with deionized water.

2.2. Microstructural analysis

The crystal structure and phase of the as-prepared materials were characterized by X-ray diffraction (XRD) using Cu K α radiation with 2 θ ranging from 20 to 90 and a scanning current 200 mA. The microstructure morphology and chemical composition were analyzed by scanning electron microscopy (SEM) using a Hitachi S-3400 scanning electron microscope with a genesis-type energy dispersive X-ray spectroscopy (EDS) attachment. Through the detection results, the element composition and element distribution of the test sample can be known, and we can intuitively obtain the element distribution and content. The composition and bonding state of surface layers formed on the coatings were determined using X-ray photoelectron spectroscopy (XPS). An ESCALAB 250Xi

system that employed an Al K α X-ray source was used. Spectra were calibrated with respect to adventitious carbon (the C 1 s peak) at a binding energy of 284.6 eV. Peaks from survey scans and regional scans were identified with reference to a standard XPS database.

2.3. Hardness and wear

Thickness of the coatings with an electronic spiral micrometer, each coatings is measured five times and averaged. The HV-1000IS microhardness tester was used to test the Vickers hardness of the coatings. The load was 50 N and lasted 15 s. Each sample is tested five times, and the average value of the five test results is taken for comparative analysis. The HSR-2 M high-speed reciprocating friction and wear tester was used to test the friction resistance of the coatings. Through this test, the friction coefficient, wear amount and wear pattern of the coatings can be obtained. The experimental process is that the applied load is 3 N, the friction time is 30 min, the speed is 200 r/min, the rotating friction is one-way rotating friction, and it is carried out at room temperature.

2.4. Corrosion behavior measurement

Electrochemical methods are usually used to evaluate the corrosion behavior of materials. The potentiodynamic polarization curves of HEA coatings and Cu substrate were performed on an Interface1000 electrochemical workstation using a typical three-electrode system in 3.5 wt% NaCl solution. The reference electrode was a saturated calomel electrode, the counter electrode was a platinum electrode, and the sample was a working electrode. The corrosive medium is 3.5 wt% NaCl solution, and the scanning speed is 10 mV/s.

3. Results and discussion

3.1. Microstructure analysis

Fig. 1 shows the SEM of FeCoNiCr, FeCoNiCrMn, FeCoNiCrAl high-entropy alloy coatings. The surfaces of HEA coatings are uniform and smoothness. Fig. 2 and Table 2 shows the distribution of each element in the HEA coatings. The energy spectrum analysis results on the surface of the coatings show that Fe, Co, Ni, Cr, Mn, and Al are evenly distributed.

Fig. 3 shows the XRD patterns of FeCoNiCr, FeCoNiCrMn, FeCoNiCrAl high-entropy alloy coatings. It can be seen from Fig. 3 that the diffraction peak of the high-entropy alloy coatings broadens around $2\theta = 45^\circ$, and all elements are dissolved into the same crystal lattice to form a single disordered solid solution phase. The dynamic hysteresis diffusion effect of high-entropy alloys inhibits the diffusion in the alloy, making it difficult for atoms to diffuse and increasing the solid solubility between different alloying elements, thereby increasing the lattice parameters. At the same time, the difference in atomic size causes distortion of the crystal lattice and also increases the lattice parameters. At this time, the coatings is an amorphous coatings. The atomic radii of Al and Mn are slightly larger, which causes the lattice expansion around the solute atoms to cause the amorphous diffraction peak to shift to the left. The FCC phase diffraction peak is observed in Fig. 3. As shown in Fig. 3, it was noticed that the intensity of the FCC phase diffraction peak became stronger. This was due to Al element playing an FCC stabilizer in the composition design of the HEAs [30]. While the Al element is added, the

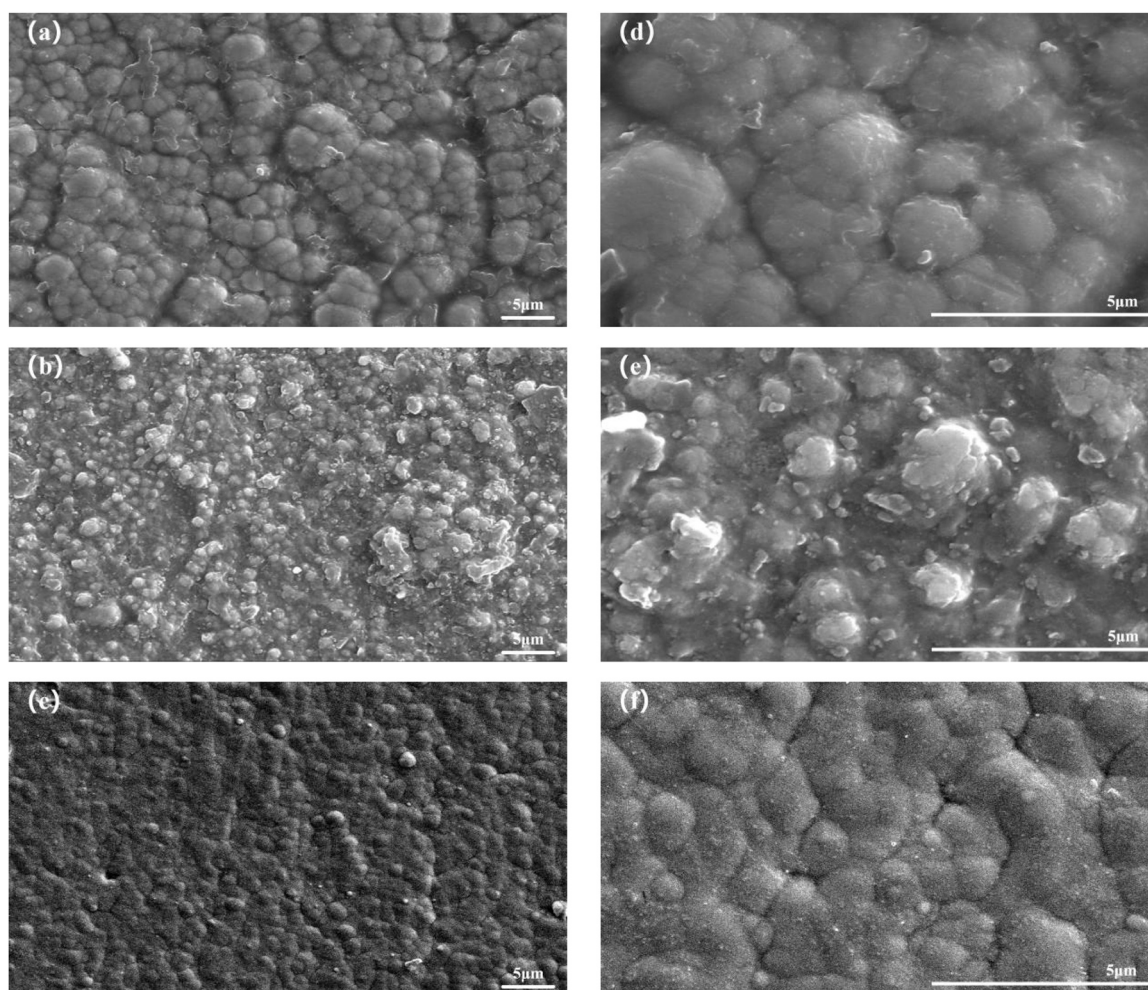


Fig. 1. Microstructure of FeCoNiCrX (X=Mn, Al) HEA coatings. FeCoNiCr HEA coatings (a, d), FeCoNiCrMn HEA coatings (b, e), FeCoNiCrAl HEA coatings(c, f).

Al element promoted the generation of FCC phase. The XRD analysis result is compared with the standard PDF card, and it is found that the FCC phase diffraction peak is determined to be the (Fe, Ni) solid solution space group Fm $\bar{3}m$.

The surface of FeCoNiCr, FeCoNiCrMn, and FeCoNiCrAl HEA coatings were examined by XPS to determine the coatings composition, as shown in Figs. 4–6. The XPS survey spectrum confirmed that the presence of O, Al, Co, Cr, Fe, Ni, and Mn elements in the surface. From Fig. 4(b), the binding energy of element C in the coatings was 284.68 eV, which was extremely close to the standard binding energy of element C (284.6 eV). This manifested the accuracy of the test results of XPS. Further, the corresponding high-resolution spectrum was used to detect the chemical valence of all elements. The narrow scan results of Co $2p_{3/2}$, Cr $2p_{3/2}$, Fe $2p_{3/2}$, Ni $2p_{3/2}$, Al $2p_{3/2}$, and Mn $2p_{3/2}$ after shift correction are shown in Figs. 4–6. Table 3 [31–33] lists the corresponding binding energies of the peaks. The chemical composition of the surface was determined by fitting the high-resolution spectrum of each element. Partial oxides can be detected in these high-resolution spectra.

As shown in Figs. 4–6, the Fe $2p_{3/2}$ spectra were split into three constituent peaks representing the metallic Fe, Fe $_2O_3$, and Fe $_3O_4$

oxide species. The Co $2p_{3/2}$ spectra are split into constituent peaks, namely, Co $_3O_4$ and metal CoO. Ni $2p_{3/2}$ were split into metallic Ni and Ni $^{2+}$ (NiO/Ni(OH) $_2$). Al $2p_{3/2}$ was separated into two strong peaks of metallic Al and Al $_2O_3$. The Cr $2p_{3/2}$ spectra were split into the metallic Cr $_2O_3$ species. The Mn $2p_{3/2}$ spectra are split into constituent peaks, namely, Mn $_2O_3$. For alloys the stoichiometry is particularly important [26–29]. These primary oxides in the HEA coatings may have been formed in air or in a water bath, where the surface elements on the coatings are oxidised.

3.2. Hardness of the coatings

The thicknesses of the FeCoNiCr, FeCoNiCrMn, and FeCoNiCrAl HEA coatings are $54 \pm 0.2 \mu m$, $61 \pm 0.1 \mu m$, $57 \pm 0.8 \mu m$. Fig. 7 shows the hardness of substrate ($158.0 \pm 10 Hv$), FeCoNiCr ($593.4 \pm 15 Hv$), FeCoNiCrMn ($657.2 \pm 12.8 Hv$), and FeCoNiCrAl ($647.4 \pm 11.7 Hv$) high-entropy alloy coatings. The hardness of the coatings is higher than that of the substrate. The addition of Cr elements significantly improves the hardness of the coatings, which is mainly due to the fact that Cr elements are solidly soluble in the FCC phase of the coatings, resulting in greater lattice distortion

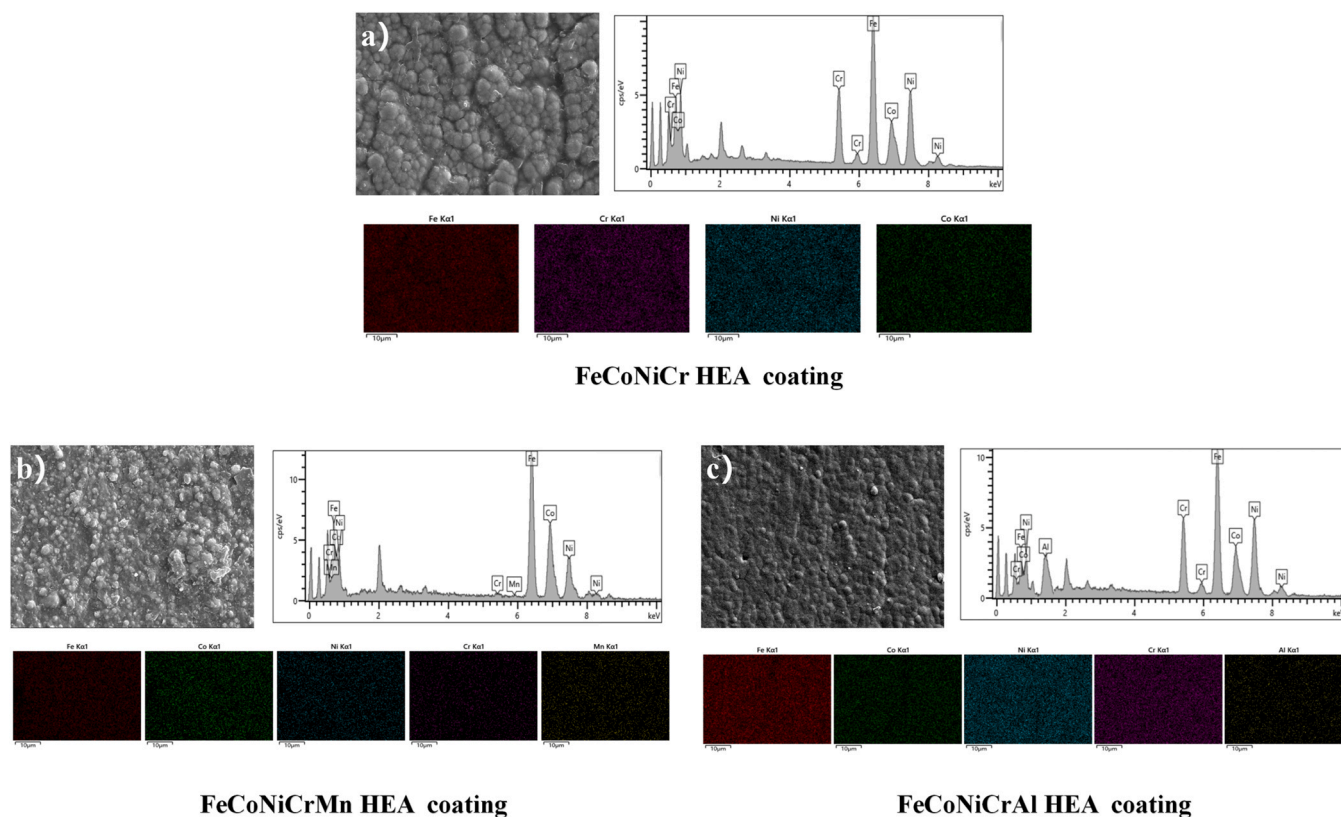


Fig. 2. Distribution of each element in the HEA coatings.

Table 2
Average elemental composition of HEA coatings obtained from SEM-EDS analysis.

Elements/ System	FeCoNiCr HEA coatings	FeCoNiCrMn HEA coatings	FeCoNiCrAl HEA coatings
Fe	21.60	23.19	23.56
Co	19.39	20.38	15.32
Ni	30.52	20.48	26.64
Cr	28.49	15.51	14.38
Mn	–	20.44	–
Al	–	–	20.10

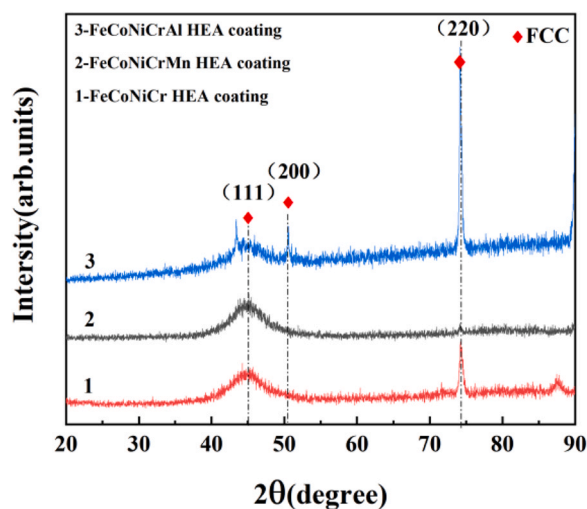


Fig. 3. XRD patterns of FeCoNiCr (X=Mn, Al) HEA coatings.

effect, which increases the hardness of the FCC phase. As can be seen in Fig. 1, in the FeCoNiCrMn and FeCoNiCrAl HEA coatings, the grains are uniformly refined, resulting in an increase in hardness.

3.3. Tribological properties of the coatings

Fig. 8 shows the friction coefficient and wear mass loss of FeCoNiCr, FeCoNiCrMn, and FeCoNiCrAl high-entropy alloy coatings and substrates. It is observed in the Fig. 8, that the friction coefficient of the high-entropy alloy coatings is significantly lower than that of the substrate, and the wear mass is also reduced. It indicates that the coatings has higher wear resistance than the substrate. The hardness of the material and the wear resistance are directly proportional, and the coatings has a higher hardness than the substrate, so the coatings has a better wear resistance.

Fig. 9 shows the wear morphology of the coatings. It can be seen from Fig. 9a that plough furrows appear after the coatings is worn. And Fig. 9d shows that the coatings has adhesion, indicating that a small amount of material transfer occurs on the FeCoNiCr high-entropy alloy coatings after wear, and some of the wear debris with higher hardness is in the coatings. It plays the role of abrasive particles between the pair of grinding pairs, causing scratches and micro-cutting on the surface of the coatings. In Fig. 9b, it is observed that the abraded coatings is smooth with small cracks, and in Fig. 9e, it can be seen that the coatings falls off and produces more fine particles. This is because the FeCoNiCrMn coatings is subjected to cyclic contact stress in a local area during the abrasion process, and the repeated deformation caused the crack to generate and propagate. The coatings peeled off to form wear debris, and some of the wear debris with lower hardness adhered to the coatings due to repeated rolling. It can be seen from Fig. 9c that the coatings has

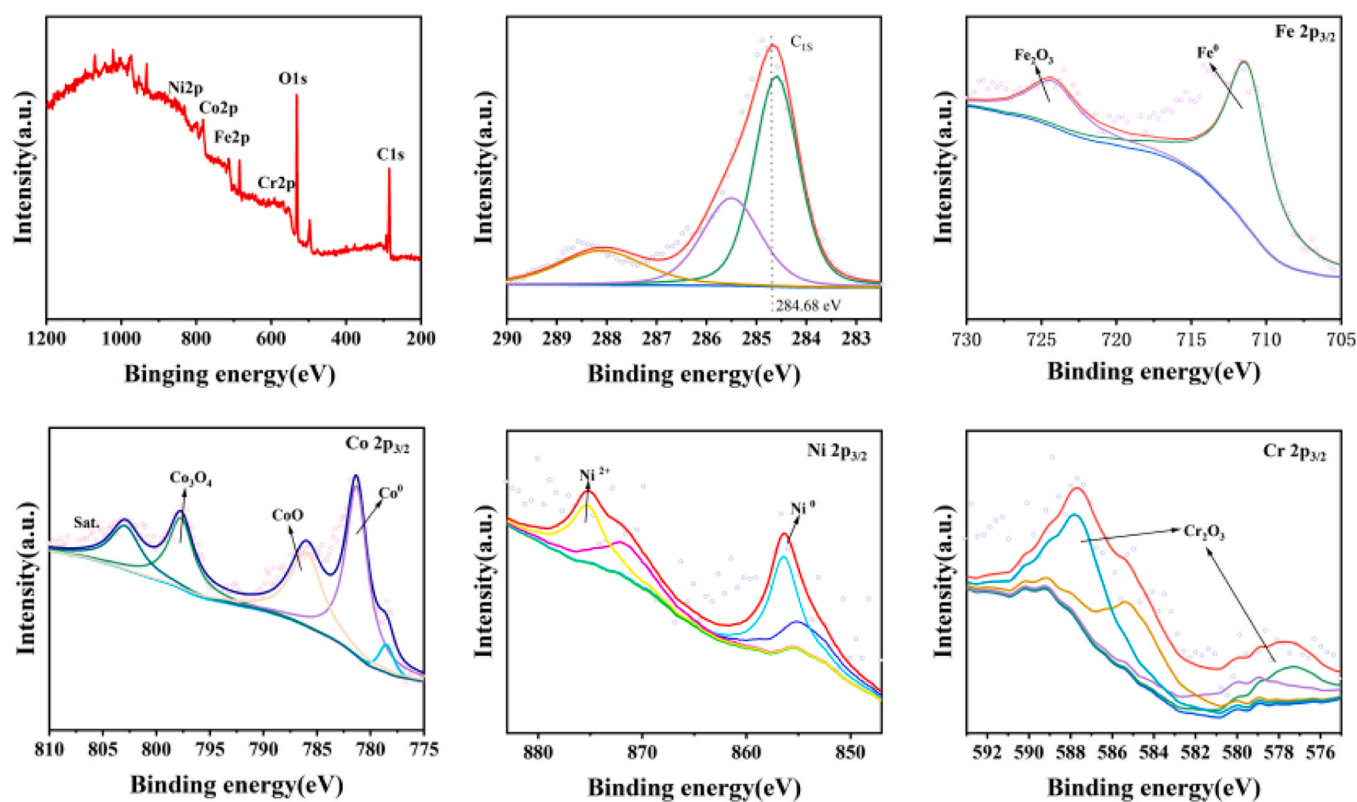


Fig. 4. Deconvoluted Fe 2p_{3/2}, Co 2p_{3/2}, Ni 2p_{3/2}, Cr 2p_{3/2}, and XPS survey of FeCoNiCr HEA coatings.

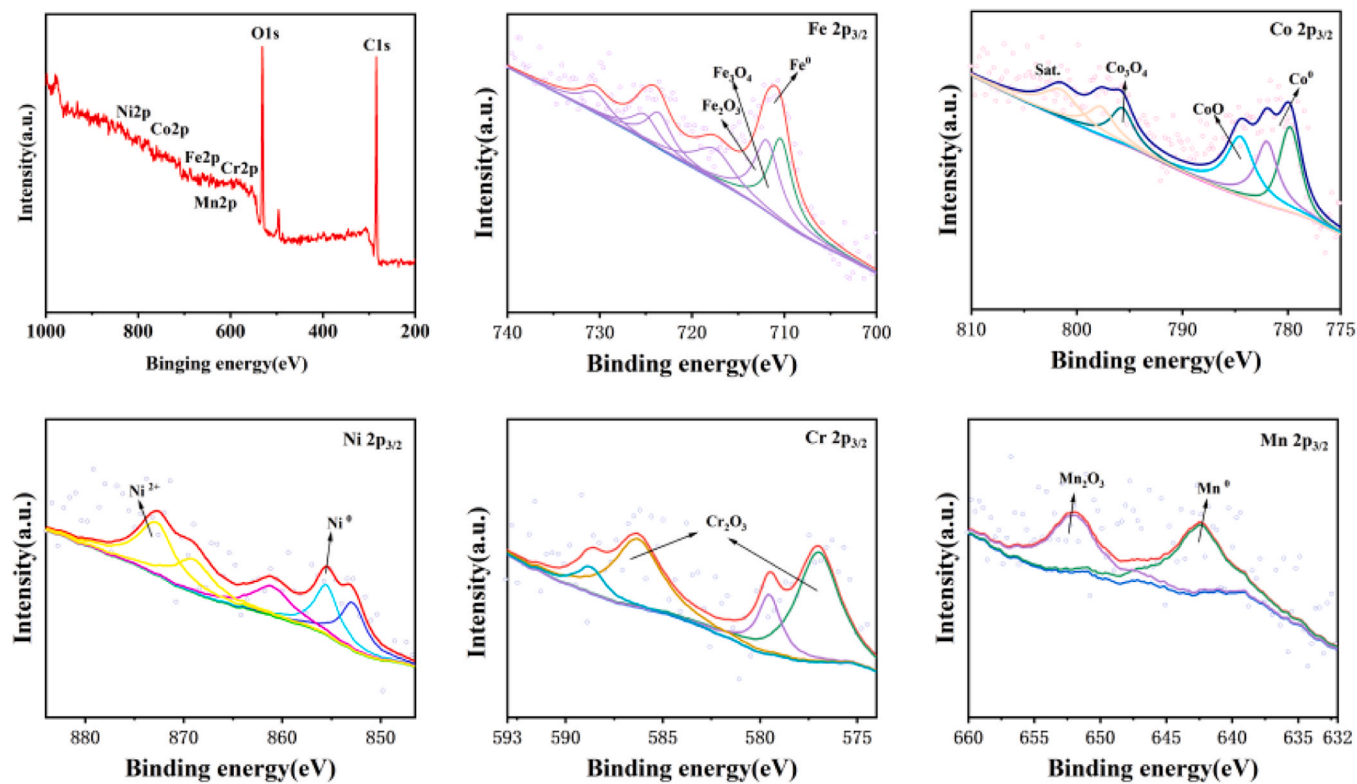


Fig. 5. Deconvoluted Fe 2p_{3/2}, Co 2p_{3/2}, Ni 2p_{3/2}, Cr 2p_{3/2}, Mn 2p_{3/2}, and XPS survey of FeCoNiCr HEA coatings.

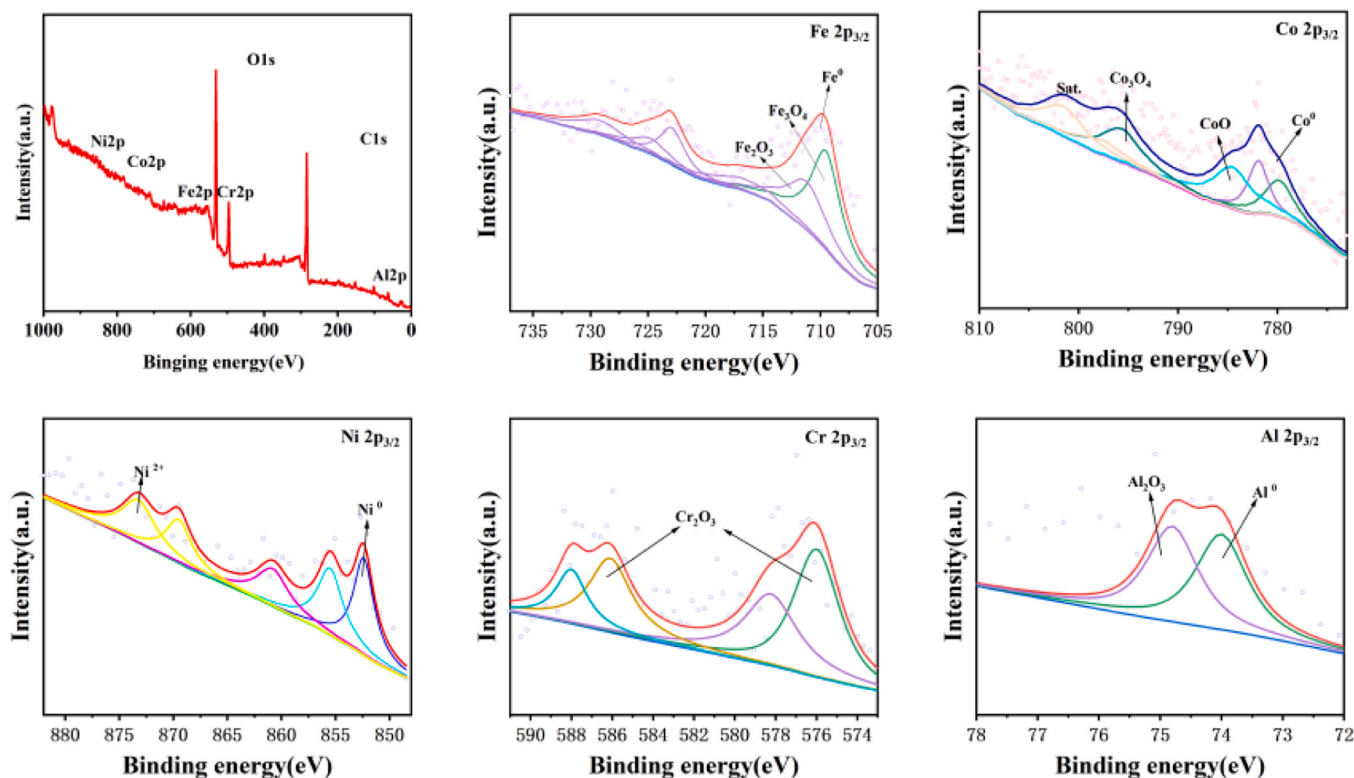


Fig. 6. Deconvoluted Fe 2p_{3/2}, Co 2p_{3/2}, Ni 2p_{3/2}, Cr 2p_{3/2}, Al 2p_{3/2}, and XPS survey of FeCoNiCrAl HEA coatings.

Table 3

Binding energies of Al 2p_{3/2}, Cr 2p_{3/2}, Ni 2p_{3/2}, Co 2p_{3/2}, Fe 2p_{3/2}, and Mn 2p_{3/2} by XPS analysis of the surface in FeCoNiCrX (X=Mn, Al) HEA coatings [31–33].

Elements	FeCoNiCr HEA coatings	FeCoNiCrMn HEA coatings	FeCoNiCrAl HEA coatings	Peak
Fe	2p _{3/2}	2p _{3/2}	2p _{3/2}	Fe ⁰
	724.8	711.98	711.78	Fe ₂ O ₃
Co	2p _{3/2}	2p _{3/2}	2p _{3/2}	Fe ₃ O ₄
	786.18	784.38	784.78	Co ⁰
	797.78	795.68	795.48	CoO
				Co ₃ O ₄
Ni	2p _{3/2}	2p _{3/2}	2p _{3/2}	Ni ⁰
	873.98	872.88	873.28	NiO/
Cr	2p _{3/2}	2p _{3/2}	2p _{3/2}	Ni(OH) ₂
	577.68	577.08	576.78	Cr ⁰
	587.78	586.38	586.08	Cr ₂ O ₃
				Cr ₂ O ₃
Mn	2p _{3/2}	2p _{3/2}	2p _{3/2}	Mn ⁰
		651.88		Mn ₂ O ₃
Al	2p _{3/2}	2p _{3/2}	2p _{3/2}	Al ⁰
			74.78	Al ₂ O ₃

peeling pits, and in Fig. 9d, it can be seen that the coatings has adhesion. Therefore, the failure mode of FeCoNiCrAl coatings is mainly local peeling, accompanied by a small amount of adhesive wear. In terms of the width of the abrasion marks, the FeCoNiCrMn HEA coatings and the FeCoNiCrAl HEA coatings have a smaller width compared to the FeCoNiCr HEA coatings, indicating that the FeCoNiCrMn HEA coatings and the FeCoNiCrAl HEA coatings may have better wear resistance.

3.4. Corrosion performance of the coatings

The corrosion behaviour of high entropy alloy coatings in 3.5 wt% NaCl solutions is mainly accomplished by electron transfer, which

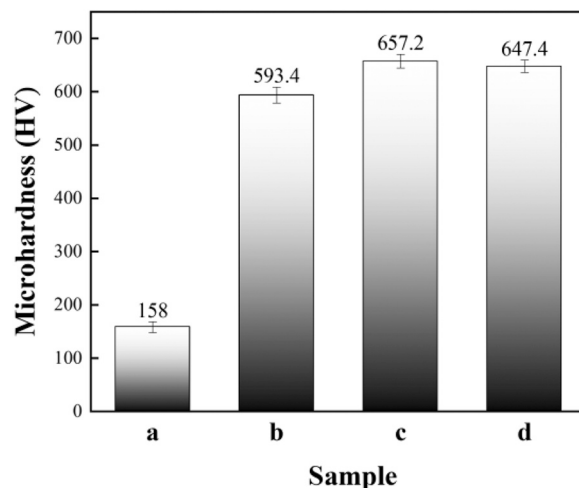


Fig. 7. The microhardness distribution of FeCoNiCrX (X=Mn, Al) HEA coatings and substrate. a) Cu substrate, b) FeCoNiCr HEA coatings, c) FeCoNiCrMn HEA coatings, d) FeCoNiCrAl HEA coatings.

consists of anodic dissolution and cathodic reduction. Anodic dissolution occurs when metal ions enter the NaCl solution and leave electrons in the metal.

The net reaction: $\text{HEA} + \frac{n}{4}\text{O}_2 + \frac{n}{2}\text{H}_2\text{O} \rightarrow \text{nOH/HEA}$

During the corrosion process of HEA coatings, the oxide layer on the coatings surface consists of Fe, Co, Ni, Cr, Al and Mn oxides, which determine the extent of the degradation of the coatings. Fig. 10 shows the polarization curves of coatings and substrates in 3.5 % NaCl solution. The polarization curve of the substrate in the

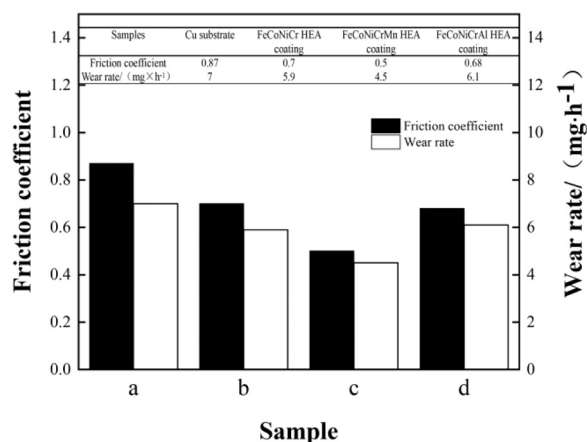


Fig. 8. Friction coefficient and wear rate of FeCoNiCrX (X=Mn, Al) HEA coatings and substrate. a) Cu substrate, b) FeCoNiCr HEA coatings, c) FeCoNiCrMn HEA coatings, d) FeCoNiCrAl HEA coatings.

passivation zone showed obvious fluctuations, indicating that the stability of the passivation film of the substrate was low and the protection ability of the substrate was poor. According to the theory of corrosion electrochemistry, the corrosion resistance of metals is negatively correlated with the self-corrosion potential, and the corrosion rate is positively correlated with the self-corrosion current

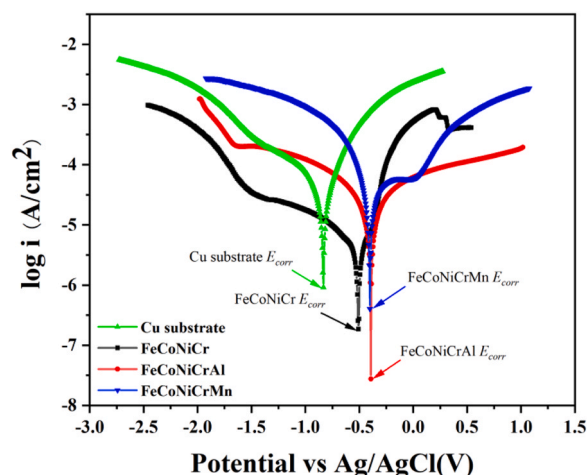


Fig. 10. Polarization curves of FeCoNiCrX (X=Mn, Al) HEA coatings and substrate.

density. The larger the E_{corr} value, the better the corrosion resistance of the metal, and the smaller the I_{corr} value, the lower the corrosion rate of the metal. Table 4 shows that the E_{corr} of the coatings is much higher than that of the substrate, and the I_{corr} is lower than that of the substrate. This indicates that the coatings is less prone to corrosion in 3.5 % NaCl solution, and its corrosion rate is much lower than that of the substrate.

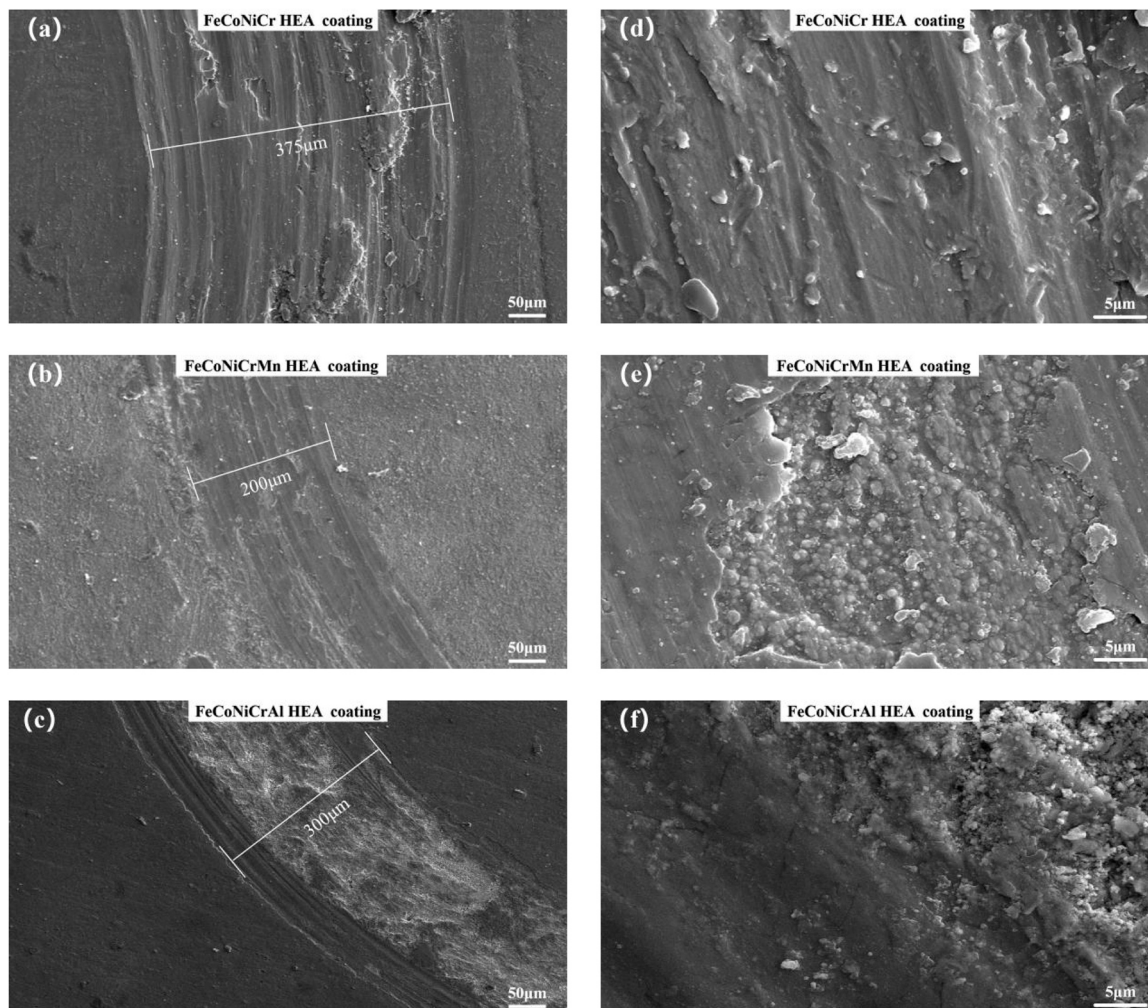


Fig. 9. Wear morphology of FeCoNiCrX (X=Mn, Al) HEA coatings. FeCoNiCr HEA coatings (a, d), FeCoNiCrMn HEA coatings (b, e), FeCoNiCrAl HEA coatings (c, f).

Table 4

Electrochemical parameters of Cu substrate, and FeCoNiCrX (X=Mn, Al) HEA coatings in 3.5 wt% NaCl solution.

Material	Cu substrate	FeCoNiCr HEA coatings	FeCoNiCrMn HEA coatings	FeCoNiCrAl HEA coatings
E_{corr}/V	-0.831 ± 0.001	-0.508 ± 0.001	-0.402 ± 0.002	-0.392 ± 0.001
$i_{\text{corr}}/\text{A cm}^{-2}$	$6.252 \times 10^{-5} \pm 0.16$	$3.327 \times 10^{-5} \pm 0.21$	$5.794 \times 10^{-5} \pm 0.11$	$1.772 \times 10^{-5} \pm 0.16$

4. Conclusion

- (1) The FeCoNiCrX (XMn, Al) high-entropy alloy (HEA) coatings were successfully prepared by the electrodeposition method, and the coatings was well bonded to the substrate. The surface microstructures of the coatings show typical cell-like structure. This high-entropy alloy coatings is a single FCC phase.
- (2) The FeCoNiCrX (Mn, Al) HEA coatings improves the microhardness and wear resistance of the substrate. The microhardness of the HEA coatings shows a 4-fold increase compared to that of the substrate material. The wear mechanisms of FeCoNiCrX (Mn, Al) HEA coatings are a combination of abrasion and adhesion wear, and the wear rate of coatings is reduced, indicating that the coatings have better wear resistance.
- (3) In 3.5 % NaCl solution, the self-corrosion potential and self-corrosion current density of substrate, FeCoNiCr HEA coatings, FeCoNiCrMn HEA coatings and FeCoNiCrAl HEA coatings are $-0.831 \pm 0.001 \text{ V}$, $6.252 \times 10^{-5} \pm 0.16 \text{ A cm}^{-2}$; $-0.508 \pm 0.001 \text{ V}$, $3.327 \times 10^{-5} \pm 0.21 \text{ A cm}^{-2}$; $-0.402 \pm 0.002 \text{ V}$, $5.794 \times 10^{-5} \pm 0.11 \text{ A cm}^{-2}$; $-0.392 \pm 0.001 \text{ V}$, $1.772 \times 10^{-5} \pm 0.16 \text{ A cm}^{-2}$. These results show that it is effective to strengthen the corrosion resistance of copper substrate by HEA coatings.

CRedit authorship contribution statement

Zhiyi Rong: Writing – original draft, Conceptualization, Data curation. **Chaohui Wang:** Conceptualization, Writing – review & editing, Supervision. **You Wang:** Validation, Supervision. **Meiling Dong:** Data curation. **Yuan You:** Data curation. **Jianing Wang:** Visualization. **Huinan Liu:** Visualization. **Jiaqi Liu:** Resources. **Yuhang Wang:** Investigation. **Zhongyu Zhu:** Investigation.

Data Availability

The data that has been used is confidential.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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